Morgan E. Koch

Erie, PA || (717) 601-4034 || ekoch.morgan@gmail.com || www.linkedin.com/in/moekoch || https://moekoch.xyz/

EDUCATION

The Pennsylvania State University *Erie, PA*
School of Engineering at Penn State Behrend *Graduation: May 2027*
Bachelor of Science in Computer Science *GPA: 3.37*
Minors: Management Information Systems (MIS) & Biology

Certification: NVIDIA Fundamentals of Deep Learning (2024)
Relevant Coursework: Object-Oriented Programming, Machine Learning, Software Engineering, Data Structures & Algorithms

RELEVANT EXPERIENCE

App Development Manager / Software Engineer (Excelerate Internship) *Erie, PA*
Virtual/Augmented Reality (VAR) Lab at Penn State Behrend *March 2025 – Present*

- Engineered and deployed Landing Place, an AI-enabled React Native/Expo mobile application (iOS) for clinical research and outcome tracking.
- Revived a non-functional legacy codebase and led a five-person team to deliver a production-ready application within one academic year.
- Owned the end-to-end release lifecycle, including build automation, versioning, provisioning, and TestFlight distribution for internal testing and active clinical trials deployed to ~80 participants.
- Established technical direction for an inherited codebase in a high-ambiguity environment and collaborated with researchers to define requirements and resolve system issues.

IT Product Management Intern *Enola, PA*
Members 1st Federal Credit Union *May 2026 – July 2026*

- Defined product and technical requirements for AI-powered chat and smart summary features within a digital banking application to enable secure, AI-driven member interactions while ensuring compatibility with legacy backend systems and regulatory requirements.
- Analyzed Azure, LlamaIndex, and LangChain architectures to support engineering decisions for RAG and agent-based AI workflows, documenting trade-offs between retrieval-heavy and agentic approaches.
- Conducted competitive analysis of AI fintech solutions, documenting recommendations for scalability, compliance, auditability, and responsible AI practices.

IT Project Management Intern *Harrisburg, PA*
TE Connectivity *May 2025 – August 2025*

- Supported cross-functional stakeholder alignment, project documentation (CONOPS, BRDs, RASCI), and global multi-region coordination across IT initiatives.

PROJECTS

Wabtec Software Plugin for Electrical Data Integration (Senior Capstone) *Erie, PA*
Penn State Behrend School of Engineering & Wabtec *Sept 2026 – Present*

- Developing a Java/Python integration tool to extract, normalize, validate, and synchronize electrical design data between MagicDraw/Teamwork Cloud and Zuken E3.series using Open APIs, XML/JSON processing, and change-detection workflows

RedScare — 1st Place Winning Hackathon Project *Erie, PA*
Penn State Behrend School of Engineering *April 11, 2026*

- Built a women's health web app in 12 hours, implementing cycle and symptom tracking, data visualizations, and AI-powered insights using JavaScript, HTML/CSS, Supabase, and the Anthropic API.
- Won first place in Penn State Behrend's Open Innovation hackathon category; subsequently selected for funded development into a Flutter mobile app with research-backed educational content.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C, C++, SQL, HTML5/CSS
Frameworks & Libraries: React Native, Expo, EJS, WordPress
Developer Tools: Git, GitHub, Docker, Figma, Firebase, Supabase
IT Support: ServiceNow, KillDisk, device imaging/repair
Design: Adobe Creative Suite, Blender (3D animation, basic rigging)

LEADERSHIP & ACTIVITIES

Vice President, Association for Computing Machinery (ACM) - Technical workshops and hackathon preparation.
Media Board & Competitive Team Member, Behrend eSports **-** Web and branding initiatives; collegiate competition.